

Notes on Kinematic Fit

The template for this note is provided here[2] [1]
October 22, 2026

[1] Bono University

Outline

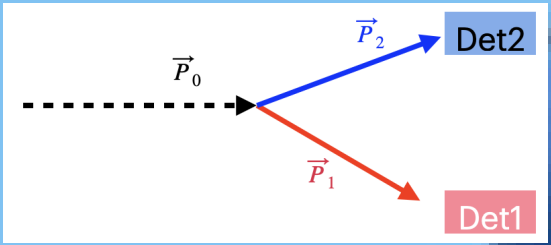
1. Introduction
2. Examples
3. Covariance Tricks
4. Applications: Ξ decay at J-PARC
5. Appendix



Introduction



Measurement Error



Consider a particle with known momentum $\vec{P}_0$ decaying into two particles with momenta $\vec{P}_1$ and $\vec{P}_2$. Detector resolution causes measured momenta to deviate from their true values, breaking the momentum conservation for the the measured momentum sum.

$$\vec{P}_0 = \vec{P}_1 + \vec{P}_2; \quad \vec{P}_{1,meas} + \vec{P}_{2,meas} \neq \vec{P}_0 \qquad (1)$$

For simplicity, consider one Cartesian momentum component, and assume independent measurement errors.

$$\chi^2 = \frac{(P_1 - P_{1,meas})^2}{\sigma_1^2} + \frac{(P_2 - P_{2,meas})^2}{\sigma_2^2} \qquad (2)$$

However, this expression is not that meaningful in practice: minimizing this expression simply returns the measured values.



Constrained Optimization with The Lagrange Multiplier

By incorporating the *Constraints*, specifically *momentum conservation*, we involve additional knowledge to (2). This is known as the *Lagrange Multiplier* method. Define the *Lagrangian* $\mathcal{L}$ as:.

$$\mathcal{L} = \frac{(P_{1,KF} - P_{1,meas})^2}{\sigma_1^2} + \frac{(P_{2,KF} - P_{2,meas})^2}{\sigma_2^2} + 2\lambda(\mathbf{P_{1,KF}} + \mathbf{P_{2,KF}} - \mathbf{P_0}) \tag{3}$$

Now we have meaningful expressions to minimize χ^2, hence get better estimations for the measurement.

$$\frac{1}{2}\frac{\partial \mathcal{L}}{\partial P_{1,KF}} = \frac{(P_{1,KF} - P_{1,meas})}{\sigma_1^2} + \lambda = 0 \tag{4}$$

$$\frac{1}{2}\frac{\partial \mathcal{L}}{\partial P_{2,KF}} = \frac{(P_{2,KF} - P_{2,meas})}{\sigma_2^2} + \lambda = 0 \tag{5}$$

$$\frac{1}{2}\frac{\partial \mathcal{L}}{\partial \lambda} = (P_{1,KF} + P_{2,KF} - P_0) = 0 \tag{6}$$



Why Does the Resolution Improve?

By solving the equations 4,5,6 and defining $\delta_i = P_{i,meas} - P_i$, we obtain the following expressions:

$$\lambda = \frac{P_{1,meas} + P_{2,meas} - P_0}{\sigma_1^2 + \sigma_2^2} = \frac{\delta_1 + \delta_2}{\sigma_1^2 + \sigma_2^2} \tag{7}$$

$$P_{1,KF} = P_{1,meas} - \sigma_1^2 \lambda \tag{8}$$

$$P_{2,KF} = P_{2,meas} - \sigma_2^2 \lambda \tag{9}$$

$$< \delta_1 >=< \delta_2 >= 0; \quad < \delta_1 \delta_2 >= 0 \tag{10}$$

$$\delta_{1,KF} = P_{1,KF} - P_1 = \frac{\sigma_2^2 \delta_1 - \sigma_1^2 \delta_2}{\sigma_1^2 + \sigma_2^2}; \quad < \delta_{1,KF} >= 0 \tag{11}$$

$$\sigma_{1,KF}^2 =< (P_{1,KF} - P_1)^2 >= \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} < \sigma_1^2 \tag{12}$$



The Covariance After Kinematic Fit

The constraint improves the individual momentum resolutions and introduces a correlation between the fitted momenta.

$$cov(P_1, P_2)_{KF} = -\frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} \tag{13}$$

$$V = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix} \rightarrow V_{KF} = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{14}$$

- Improved momentum resolution
- Negative correlation between P_1 and P_2

The negative correlation is obvious: We constrained $P_0 = P_{1,KF} + P_{2,KF}$.



Generalization to Multi-Variables

Let $\mathbf{m^0}$ denote the measured quantities, and $\mathbf{u^0}$ the initial estimates of any unknown parameters. The fit determines $\mathbf{m}$ and $\mathbf{u}$ subject to the constraints $\mathbf{f}(\mathbf{m},\mathbf{u}) = 0$. Unknown parameters are optional.

$$\mathbf{m^0} = \{m_1^0, m_2^0 \dots m_N^0\}; \quad \mathbf{u^0} = \{u_1^0, u_2^0 \dots u_J^0\}$$
$$\mathbf{f^0} = \{f_1^0(m_1^0, m_2^0, \dots m_N^0, u_1^0, u_2^0, \dots u_J^0), f_2^0, \dots f_K^0\} \tag{15}$$

Let $\mathbf{m}^0$ denote our initial measured data, and $\mathbf{m}$ represent the current estimates of the measurementsin each iterative step, just like P_{KF}s in the previous example. The Lagrange equation (3) is generalized to:

$$\mathcal{L}(\mathbf{m}) = (\mathbf{m}^0 - \mathbf{m})^\dagger V^{-1}(\mathbf{m}^0 - \mathbf{m}) + 2\lambda^\dagger \mathbf{f}(\mathbf{m},\mathbf{u}). \tag{16}$$

Here, the Lagrange multiplier $\lambda = \{\lambda_1, \lambda_2, \dots \lambda_K\}$ is not just a number but a column vector with k elements, corresponding to each kinematic constraint in $\mathbf{f}$.



$\mathcal{L}$ Minimization

We want to solve the equation

$$\vec{\nabla}\mathcal{L} = 0 \tag{17}$$

to obtain the minimized state. The differential term are listed within three groups.

$$\nabla_{\mathbf{m}}\mathcal{L} = -2V^{-1}(\mathbf{m}^0)(\mathbf{m}^0 - \mathbf{m}) + 2\mathbf{F}_{\mathbf{m}}^{\dagger}(\mathbf{m},\mathbf{u})\lambda = 0 \tag{18}$$

$$\nabla_{\mathbf{u}}\mathcal{L} = 2\mathbf{F}_{\mathbf{u}}^{\dagger}(\mathbf{m},\mathbf{u})\lambda = 0 \tag{19}$$

$$\nabla_{\lambda}\mathcal{L} = 2\mathbf{f}(\mathbf{m},\mathbf{u}). \tag{20}$$

Here, the subscripts denote partial derivatives. i.e. $((\mathbf{F}_m)_{ki} \equiv \frac{\partial f_k}{\partial m_i})$.

We Should Define...

$\mathbf{m}$	$\mathbf{u}$	$\mathbf{f}$	$\mathbf{V}$	$\mathbf{F_m}, \mathbf{F_u}$
Measured Data	Unknown parameters	Constraints	Covariance Matrix	Derivatives



Correction Pulls

A bias or resolution miss-estimation is revealed by observing the *Pull distribution* of each measurements. However, we cannot evaluate the 'true' value of measurement, hence pull for the real data is not accessible. Instead, from the the residual $\epsilon = m - m^0$ and its variance $V(\epsilon)$, we observe the *Correction Pulls* as :

$$P(\epsilon) = \epsilon / \sqrt{V(\epsilon)} \tag{21}$$

where

$$V(\epsilon) \equiv V(m) + V(m^0) - 2Cov(m, m^0). \tag{22}$$

The variance of the fitted variables, $V(m)$, is evaluated as

$$V(m) = J_{m,m^0} V(m^0) J^{\dagger}_{m,m^0} \tag{23}$$

where J_{m,m^0} is the Jacobian for m and m^0. Detailed calculations are provided in the appendix.



Examples



Example: Energy-Momentum Balance

We extend the on-dimensional example in the introduction to the general energy-moentum conservation in $B + T \to \sum_n P_i$, where B, T and P_i are the Lorentz vector of the beam, stationary target, and the recorded particles. We define:

$$\mathbf{m} = \{P_{beam}, \theta_{beam}, \phi_{beam}, P_1, \theta_1, \phi_1 \cdots, P_n, \theta_n, \phi_n\} \tag{24}$$

and the constraints are:

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \begin{pmatrix} -P_{beam,x} + \sum_n P_{i,x} \\ -P_{beam,y} + \sum_n P_{i,y} \\ -P_{beam,z} + \sum_n P_{i,z} \\ -\sqrt{P_{beam}^2 + m_{beam}^2} + m_{target} + \sum_n \sqrt{P_i^2 + m_i^2} \end{pmatrix}. \tag{25}$$

Notably, the mass constraint is naturally implemented in the energy term. Wrong PID assignment can be revealed by observing the p-value of the kinematic fit, or the pull distribution of the scattered particles.



Example: $\bar{p}p \to \phi\phi \to 4K$ Kinematic Balance

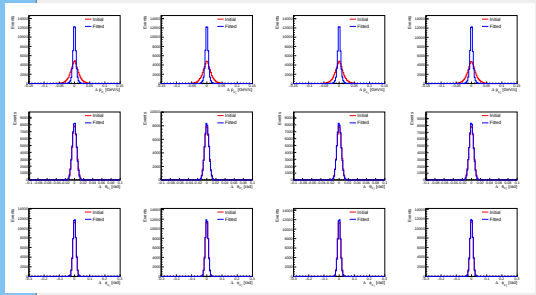


Figure: Momentum and angular residuals of the four kaons, (red) before and (blue) after the balance fit.

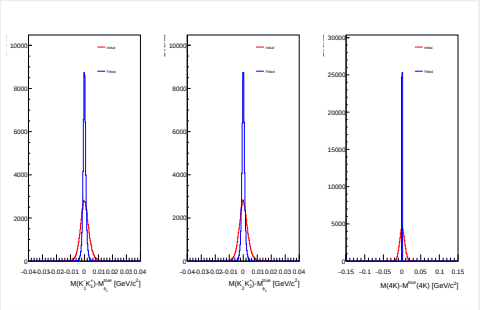


Figure: Invariant mass residuals of the two Kaon pairs, and 4K system.

Since four Kaons are emitted from the $\bar{p}p \to \phi\phi$ system, we can impose an energy-momentum balane constraint to the beam momentum and four Kaon tracks. We couduct a 4-C fit to improve the momentum and angular resolution of the Kaons, and the invariant mass resolution of the ϕ mesons.



Example: $\Lambda \to p\pi$, Defining Variables and Constraints

Assume a decay of $\Lambda \to p\pi^-$. We introduce an unknowon parameter for this example. The measurements and unknowns are defined as:

$$\mathbf{m} = \{P_p, \theta_p, \phi_p, P_\pi, \theta_\pi, \phi_\pi\}; \quad \mathbf{u} = \{P_\Lambda, \theta_\Lambda, \phi_\Lambda\} \tag{26}$$

and express the energy-momentum constraint equation as:

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \begin{pmatrix} -P_\Lambda \sin\theta_\Lambda \cos\phi_\Lambda + P_p \sin\theta_p \cos\phi_p + P_\pi \sin\theta_\pi \cos\phi_\pi \\ -P_\Lambda \sin\theta_\Lambda \sin\phi_\Lambda + P_p \sin\theta_p \sin\phi_p + P_\pi \sin\theta_\pi \sin\phi_\pi \\ -P_\Lambda \cos\theta_\Lambda + P_p \cos\theta_p + p_\pi \cos\theta_\pi \\ -\sqrt{P_\Lambda^2 + m_\Lambda^2} + \sqrt{P_p^2 + m_p^2} + \sqrt{P_\pi^2 + m_\pi^2} \end{pmatrix}. \tag{27}$$

where the mass constraint is naturally implemented in energy term.
Since we have 3 unknowns with 4 kinematic constraints, this is a 4-3 = 1-Constrained fit.



Example: $\Lambda \to p\pi$, The Derivatives

We get $\mathbf{F_u}$ and $\mathbf{F_m}$ as

$$\mathbf{F_u} = \begin{pmatrix} \frac{\partial f_1}{\partial P_\Lambda} & \frac{\partial f_1}{\partial \theta_\Lambda} & \frac{\partial f_1}{\partial \phi_\Lambda} \\ \frac{\partial f_2}{\partial P_\Lambda} & \frac{\partial f_2}{\partial \theta_\Lambda} & \frac{\partial f_2}{\partial \phi_\Lambda} \\ \frac{\partial f_3}{\partial P_\Lambda} & \frac{\partial f_3}{\partial \theta_\Lambda} & \frac{\partial f_3}{\partial \phi_\Lambda} \\ \frac{\partial f_4}{\partial P_\Lambda} & \frac{\partial f_4}{\partial \theta_\Lambda} & \frac{\partial f_4}{\partial \phi_\Lambda} \end{pmatrix}; \quad \mathbf{F_m} = \begin{pmatrix} \frac{\partial f_1}{\partial P_p} & \cdots & \frac{\partial f_1}{\partial \phi_\pi} \\ \frac{\partial f_2}{\partial P_p} & \cdots & \frac{\partial f_2}{\partial \phi_\pi} \\ \frac{\partial f_3}{\partial P_p} & \cdots & \frac{\partial f_3}{\partial \phi_\pi} \\ \frac{\partial f_4}{\partial P_p} & \cdots & \frac{\partial f_4}{\partial \phi_\pi} \end{pmatrix} \tag{28}$$

Together with the constraint equations, the fitter should evaluate these derivatives in each iteration step, and evaluate $\mathcal{L}$ using the measurement covariance matrix. The fit performance relies on the accuracy of the covariance matrix, and the χ^2 probability of the resulting $\mathcal{L}$.



Example: $\Xi \to \Lambda\pi$, $\Lambda \to p\pi$

We require two mass constraints for $\Xi \to \Lambda\pi$; $\Lambda \to p\pi$. There are some other options to define the measurements, unknowns and constraints. We choose

$$\mathbf{u} = \{P_\Xi, \theta_\Xi, \phi_\Xi\}; \quad \mathbf{m} = \{P_p, \theta_p, \phi_p, P_\pi, \theta_\pi, \phi_\pi\} \tag{29}$$

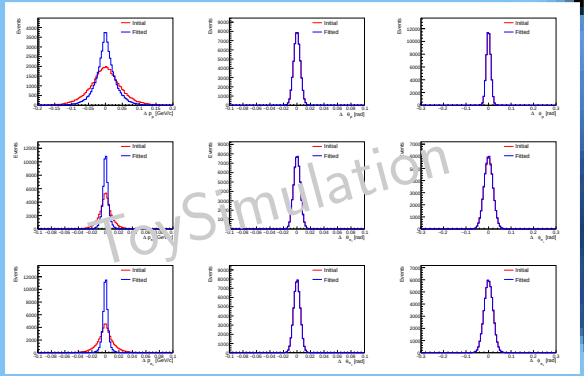
and define five constraints as:

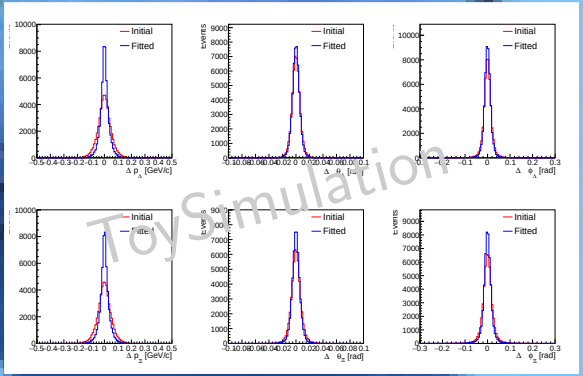
$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix} = \begin{pmatrix} -P_{\Xi,x} + P_{p,x} + P_{\pi_\Lambda,x} + P_{\pi_\Xi,x} \\ -P_{\Xi,y} + P_{p,y} + P_{\pi_\Lambda,y} + P_{\pi_\Xi,y} \\ -P_{\Xi,z} + P_{p,z} + P_{\pi_\Lambda,z} + P_{\pi_\Xi,z} \\ -E_\Lambda + E_p + E_{\pi_\Lambda} \\ -E_\Xi + E_p + E_{\pi_\Lambda} + E_{\pi_\Xi} \end{pmatrix}. \tag{30}$$

Λ variables($\{P_\Lambda, \theta_\Lambda, \phi_\Lambda\}$) are not selected in $\mathbf{u}$ to reduce matrix dimension. That is, **we don't have explicit terms** related to $\vec{P}_\Lambda$, i.e. $-P_{\Lambda,x} + P_{p,x} + P_{\pi_\Lambda,x}$ etc., because $\vec{P}_\Lambda$ are neither unmeasured nor measured variables in our choice of parameters. E_Λ in Eq. (30) is simply an abbreviation for $\sqrt{|\vec{P}_p + \vec{P}_\pi|^2 + m_\Lambda^2}$. i.e. $\partial_{P_p} E_\Lambda ! = 0$.



Example: $\rightarrow \Lambda\pi$, $\Lambda \rightarrow p\pi$ Kinematics Restoration





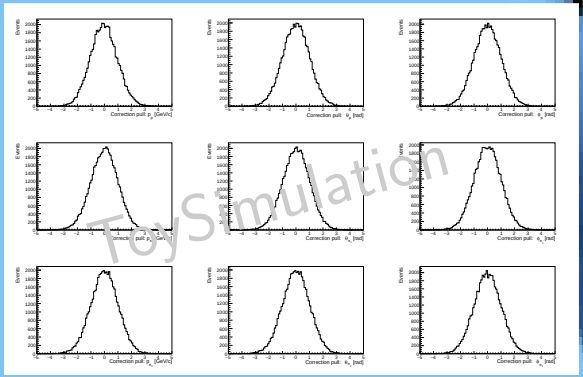
ToySimulation
ToySimulation



Covariance Tricks



Pull Distribution

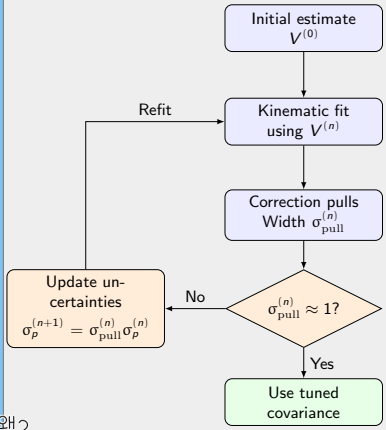


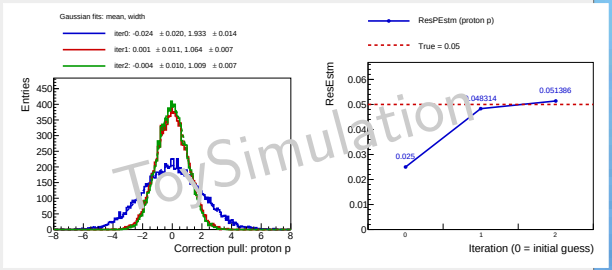
$$P(X) = \frac{X_{KF} - X_0}{\sqrt{V(X_{KF} - X_0)}} \tag{31}$$

- Pull distribution shows the normalized amount of parameter adjustment.
- Gaussian distribution with $\sigma = 1$ implies good understandings in covariance of the measurement.
- In practice, resolution can be **iteratively scaled by the pull width**.



Iterative Covariance Tuning





- Initial resolution assumption is 0.5 times the true covariance.
- Initial fit gives a pull width of 1.933, nearly 1/0.5.
- The resolution assumption is updated by the pull width, and converges to the true resolution within 2 iteration.



Covariance Normalization

$$V = \begin{pmatrix} 10^{12} & 0.9 \\ 0.9 & 10^{-12} \end{pmatrix} \rightarrow V^{-1} = ? \tag{32}$$

While taking an inverse of the variance, matrix elements with different order may be added together, leading to possible numerical unstability. We introduce a normalized covariance matrix $\tilde{V}$ and a scaling matrix S avoid floating point precision problem.

$$\tilde{V} = SVS^T; S \equiv \frac{1}{\sqrt{V_{ij}}}\delta_{ij} \rightarrow \tilde{V} = \begin{pmatrix} 1 & Cov(01)/\sigma_1\sigma_2 & \cdots \\ Cov(01)/\sigma_1\sigma_2 & 1 & \\ \cdots & & \cdots \end{pmatrix} \tag{33}$$

We can take out scaling factors in S. Measurement vectors could share the same problem, so they should also be scaled. We rewrite equation (16)

$$\chi^2 = dM^\dagger V^{-1} dM + \cdots = d\tilde{M}^\dagger \tilde{V}^{-1} d\tilde{M} + \cdots; \quad d\tilde{M} = S(M - M_0) \tag{34}$$



Off-diagonal Reduction

$$V = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \rightarrow V^{-1} = ? \tag{35}$$

Adding off-diagonal term could make matrix uninvertable. Also we require $\chi^2 = dMV^{-1}dM > 0$; V^{-1} (hence V)should be *Positive Definite*. Then, we can 'damp' the offdiagonal elements.

$$while(IsPositiveDefinite(V)) \tag{36}$$

$$V_{ij} \rightarrow V_{ij} - \alpha(1 - \delta_{ij})V_{ij} \tag{37}$$

The damping factor α should be small, e.g. 0.1, to avoid losing too much information.

Property of Positive Definite Matrix

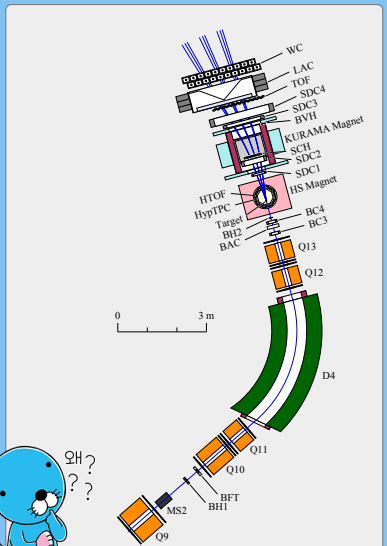
All Eigenvalues are Positive! TMatrixD well-supports eigenvalues, so we can just use it.



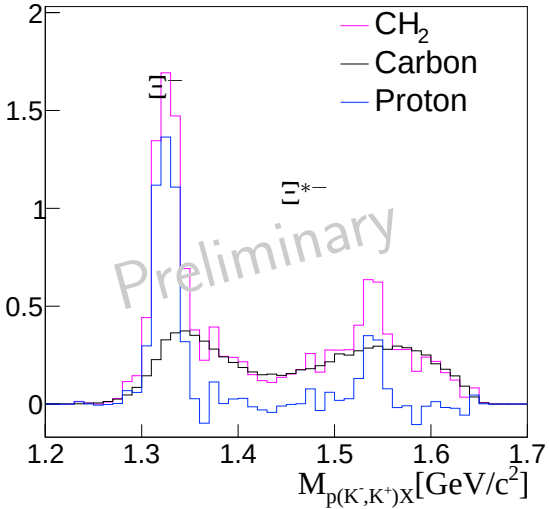
Applications: Ξ decay at J-PARC



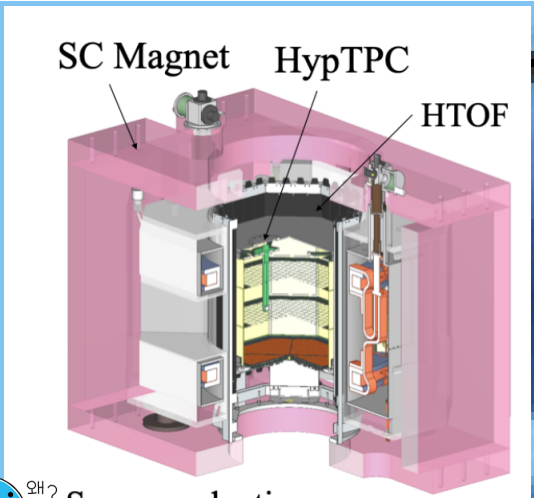
$p(K^-, K^+)X$ **at J-PARC E42**



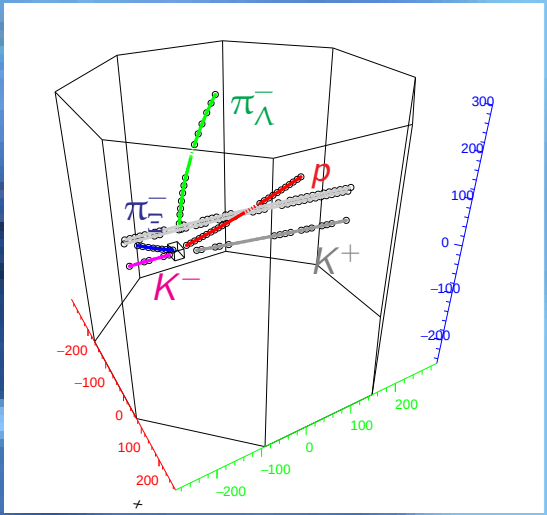




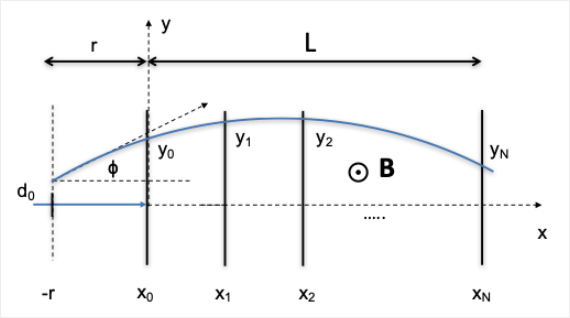
HypTPC



Superconducting
Hyperon Spectrometer



Gluckstern Formula



Z. Drasal, W. Riegler, Nucl. Instrum. Methds. A, 910, 127-132 (2018)

$$\frac{\sigma_{P_T}}{P_T} \simeq \frac{P_T}{0.3L^2B}\sqrt{\frac{720}{N+4}}\sigma_T \quad (38)$$

$$\frac{\sigma_{P_T,m.s}}{P_T} \simeq \frac{0.0136^1}{0,3\beta BL}\sqrt{\frac{d_{tot}}{X_0}} \quad (39)$$

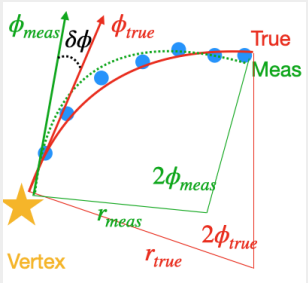
Units in GeV/c

- Momentum resolution comprises geometrical term and scattering term
- In practice, empirical rescaling factor should be multiplied

[1]G.R. Lynch and O.I Dahl, Nucl. Instrum. Methods B58, 6 (1991).



Covariance Matrix in Helix Track



- Variance in momentum modifies the curvature of the helix → direction at the vertex changes.
- 'Position' of the helix is defined from the TPC hits. → Center-of-gravity should be fixed.

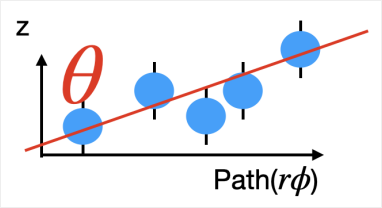
Denote the tangent angle at the center be ϕ_0 and path length to the vertex l.

$$\phi = \phi_0 \pm \frac{l}{2r}; \quad \delta\phi = \pm\frac{l}{2r}\frac{\delta r}{r} = \pm\frac{l}{r}\frac{\delta p_T}{p_T} \tag{40}$$

$$\sigma^2_\phi = \langle\delta\phi\delta\phi\rangle = \frac{l^2}{r^2 p_T^2}\sigma^2_{p_T}; \quad \mathrm{Cov}(\phi, p_T) = \langle\delta\phi\delta p_T\rangle = \pm\frac{l}{r p_T}\sigma^2_{p_T} \tag{41}$$



Covariance Matrix in Helix Track



$$h(t) : \{r\cos(\phi) - c_x, r\sin\phi - c_y, dz * r\phi - z_0\} \quad (42)$$

- The 'pitch' parameter, dz, is the slope along the circular trajectory

$\theta = \frac{\pi}{2} - \arctan(dz)$, we estimate the variance of θ based on the fitting error of dz. The error is estimated from the slope error of a linear fit:

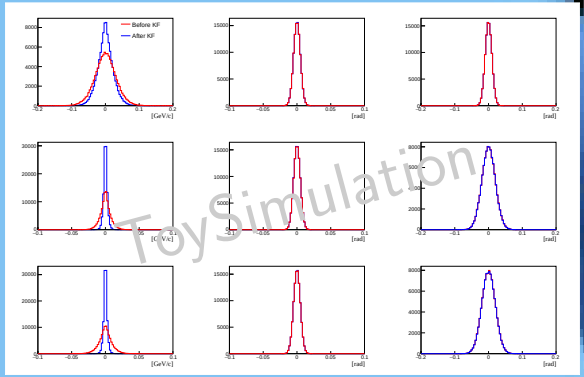
$$\sigma^2_{dz} = \frac{\sum \delta_z^2/(n-2)}{\sum (x-\bar{x})^2} \simeq \frac{n\sigma_z^2/(n-2)}{nL^2/12}; \quad \sigma_\theta = \frac{\partial dz}{\partial \theta}\sigma_{dz} = \frac{-1}{1+dz^2}\sigma_{dz}.$$

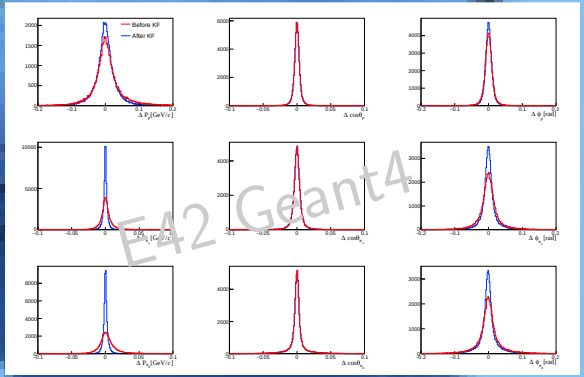
Note that, the momentum $p_z = p_T dz$ would also have some covariance with θ,

$$\langle \delta p \delta\theta \rangle = dz\langle \delta p_T \delta\theta \rangle \xrightarrow{0} + p_T \langle \delta dz \delta\theta \rangle = \frac{p_T}{1+dz^2}\sigma^2_{dz}.$$



ϕ Restoration from Diagonal Component

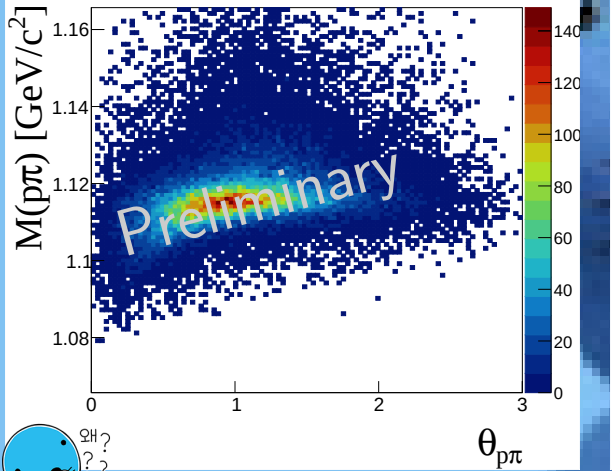


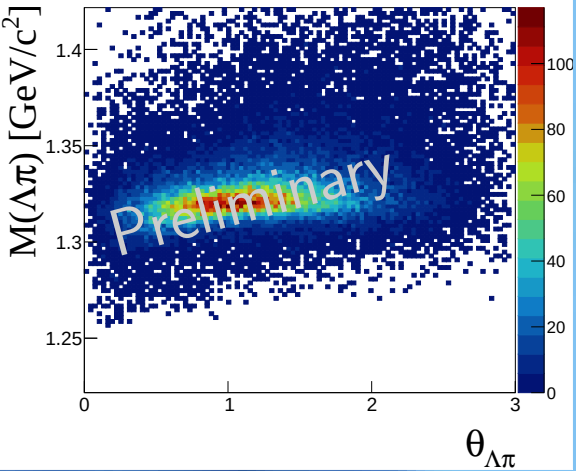


ToySimulation
E42 Geant4



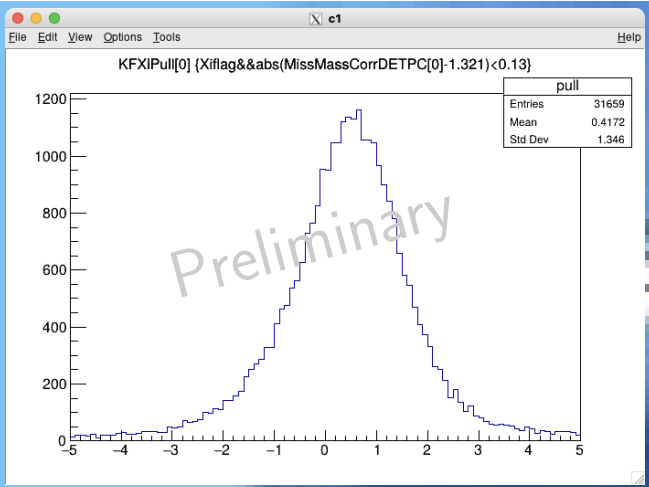
Invariant Mass Bending from Momentum Bias





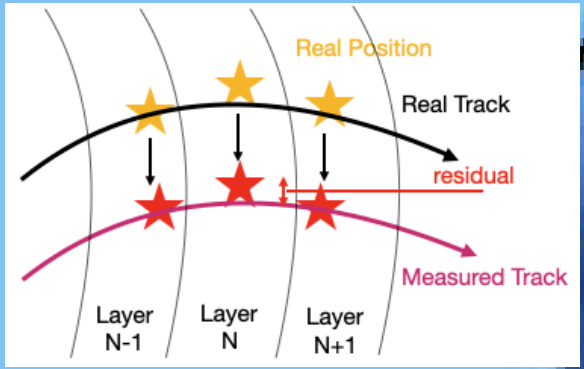


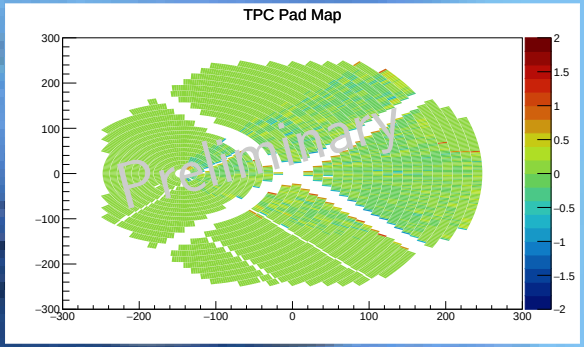
Off-center Pull Distribution





Position Residual?

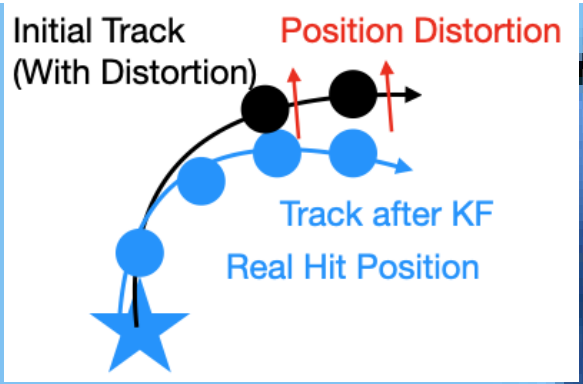


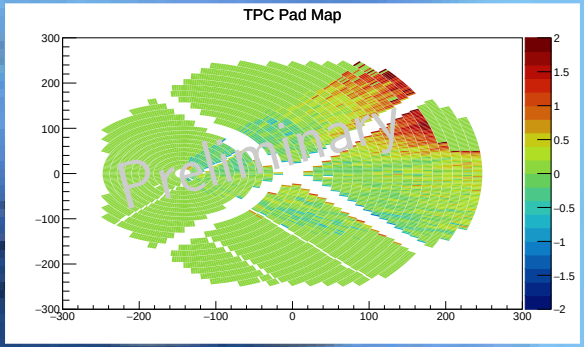


- Local, but simultaneous shift cannot be detected from position residual measurement.
- External reference for track should be provided to estimate 'true' trajectory



Position Residual from KF Track

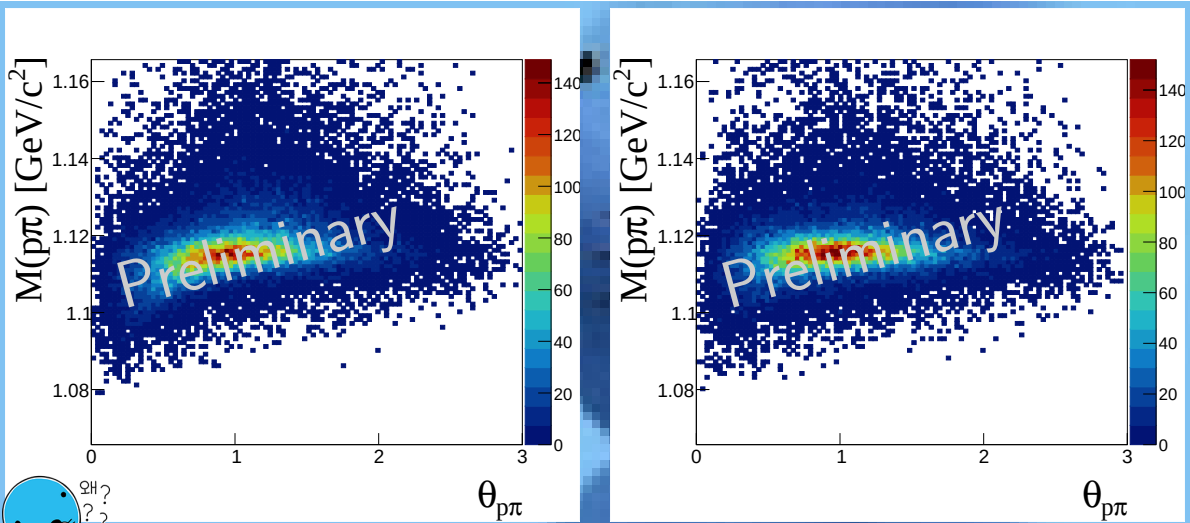




- From Kinematic fit, 'true' momentum, hence trajectory is estimated

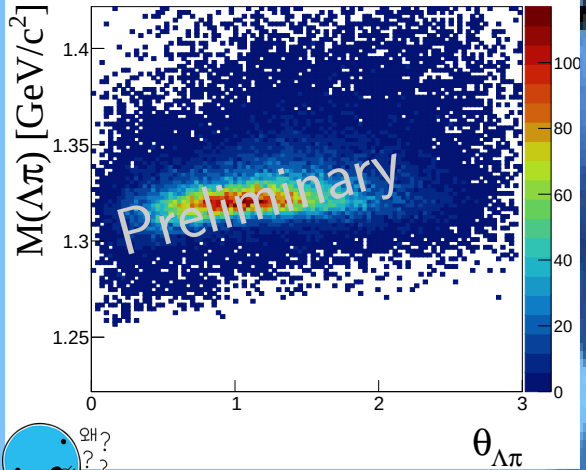


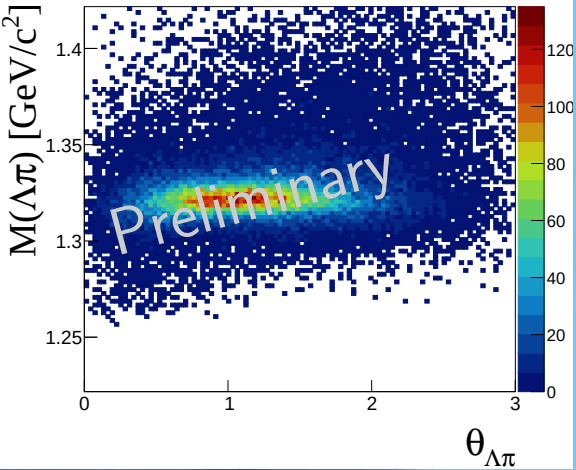
Λ After Position Correction





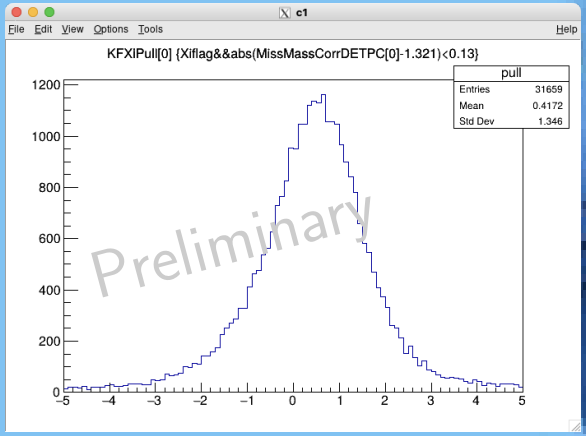
Ξ After Position Correction

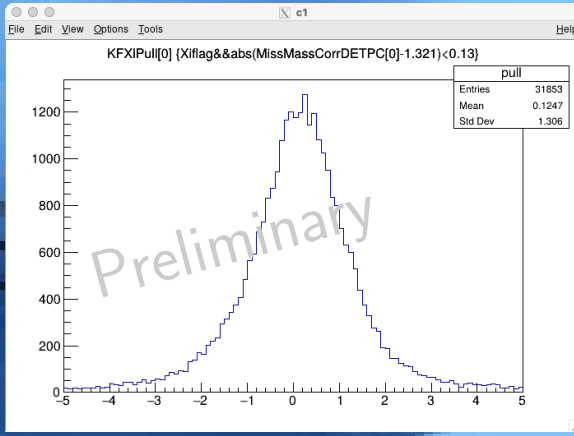






Pull distribution After Position Correction







Summary

- Measurement error can be reduced by correlating the measurements with physical constraints; This process is Kinematic Fit.
- For each kinematics, we should define a proper set of parameters, constraints and covariance matrix to perform Kinematic Fit.
- The covariance of the measurement can be tuned iteratively by the pull distribution of the Kinematic Fit.
- Additional perforance can be achieved by assigning proper off-diagonal elements in the covariance matrix.



Appendix



Minimization Steps.

The coupled differential equations (18),(20) and (19) will be solved iteratively. For each νth step,

$$V^{-1}(\mathbf{m}^0)(\mathbf{m}^{\nu+1} - \mathbf{m}^0) + (\mathbf{F}_m^\dagger)^\nu \lambda^{\nu+1} = 0 \tag{43}$$

$$(\mathbf{F}_u^\dagger)^\nu \lambda^{\nu+1} = 0 \tag{44}$$

$$\mathbf{f}^\nu + \mathbf{F}_m^\nu(\mathbf{m}^{\nu+1} - \mathbf{m}^\nu) + \mathbf{F}_u^\nu(\mathbf{u}^{\nu+1} - \mathbf{u}^\nu) = 0. \tag{45}$$

Equation (45) is not a direct consequence of Equation (19) but rather a *linear approximation*.

Note that, as we are determining the parameters $\mathbf{m}, \mathbf{u}$ and λ, they are indexed as $\nu + 1$, while constraint terms(i.e. $\mathbf{f}, \mathbf{F}_\mu$ and $\mathbf{F}_\nu$) are calculated from current step, ν. This fit is basically using Newton's method.



Coupled Equation Solving (1)

Multiplying $\mathbf{V}$ to Equation (43) leads to:

$$\mathbf{m}^{\nu+1} - \mathbf{m}^0 = -V(\mathbf{m}^0)(\mathbf{F}_m^\dagger)^\nu \lambda^{\nu+1}. \tag{46}$$

Substituting Equation (46) into Equation (45),

$$\begin{aligned}\mathbf{F}_u^\nu(\mathbf{u}^{\nu+1} - \mathbf{u}^\nu) &= -\mathbf{f}^\nu - \mathbf{F}_m^\nu(-V(\mathbf{m}^0)(\mathbf{F}_m^\dagger)^\nu \lambda^{\nu+1} + \mathbf{m}^0 - \mathbf{m}^\nu) \\ &= S\lambda^{\nu+1} - R\end{aligned} \tag{47}$$

where we define the *constraiint covariance* S and *residual* R as:

$$S \equiv \mathbf{F}_\mathbf{m}^\nu V(\mathbf{m}^0)(\mathbf{F}_\mathbf{m}^\dagger)^\nu; \quad R \equiv \mathbf{f}^\nu + \mathbf{F}_\mathbf{m}^\nu(\mathbf{m}^0 - \mathbf{m}^\nu) \tag{48}$$

Multiplying $(\mathbf{F}_u^\dagger)^\nu S^{-1}$ into (47), we get:



$$(\mathbf{F}_u^\dagger)^\nu S^{-1}\mathbf{F}_u^\nu(\mathbf{u}^{\nu+1} - \mathbf{u}^\nu) = \cancelto{0,\because (44)}{(\mathbf{F}_u^\dagger)^\nu \lambda^{\nu+1}} - (\mathbf{F}_u^\dagger)^\nu S^{-1} R. \tag{49}$$

Coupled Equation Solving (2)

Then we naturally derive the expressions

$$\mathbf{u}^{\nu+1} = \mathbf{u}^{\nu} - ((\mathbf{F}_u^{\dagger})^{\nu} S^{-1} \mathbf{F}_u^{\nu})^{-1} (\mathbf{F}_u^{\dagger})^{\nu} S^{-1} R. \tag{50}$$

and from (47),

$$\lambda^{\nu+1} = S^{-1}(\mathbf{F}_u^{\nu}(\mathbf{u}^{\nu+1} - \mathbf{u}^{\nu}) + R). \tag{51}$$

For a summary, we have obtained all equations to proceed to the next step. All other matrices in the equation can be calculated from parameters of the current step, and χ^2 can be evaluated from (16) .

$$\begin{cases} \mathbf{u}^{\nu+1} = \mathbf{u}^{\nu} - ((\mathbf{F}_u^{\dagger})^{\nu} S^{-1} \mathbf{F}_u^{\nu})^{-1} (\mathbf{F}_u^{\dagger})^{\nu} S^{-1} R & (50) \\ \lambda^{\nu+1} = S^{-1}(\mathbf{F}_u^{\nu}(\mathbf{u}^{\nu+1} - \mathbf{u}^{\nu}) + R) & (51) \\ \mathbf{m}^{\nu+1} = \mathbf{m}^{0} - V(\mathbf{m}^{0})(\mathbf{F}_m^{\dagger})^{\nu} \lambda^{\nu+1} & (46) \end{cases}$$



Covariance Matrix Propagation

We estimate the covariance for the fitted variables, m,

$$V(m) = J_{m,m^0} V(m^0) J^{\dagger}_{m,m^0} \tag{52}$$

To solve (52) the Jacobian should be determined.

$$J_{m,m^0(i,j)} = \frac{\partial m_i}{\partial m_j^0} \tag{53}$$



Jacobian

To begin with, let us express Eq (46) in terms of m^0. At the moment we will drop the superscript ν. As $\mathbf{f}(\mathbf{m}, \mathbf{u})$is a constant on m^0, $\mathbf{F_m}$ also will be a constant to m^0. Then we only need to consider the derivatives of λ. By substituting (50) ,

$$\lambda = S^{-1}(-\mathbf{F}_u(((\mathbf{F}_u^\dagger)S^{-1}\mathbf{F}_u)^{-1}(\mathbf{F}_u^\dagger)S^{-1}R) + R) \tag{54}$$

and the residual matrix is:

$$R \equiv \mathbf{f} + \mathbf{F_m}(\mathbf{m}^0 - \mathbf{m}) \rightarrow \frac{\partial R}{\partial m^0} = \mathbf{F}_m \tag{55}$$

Now we obtain the derivative of λ as:

$$\frac{\partial \lambda}{\partial m^0} = S^{-1}(-\mathbf{F}_u((\mathbf{F}_u^\dagger S^{-1}\mathbf{F}_u)^{-1}\mathbf{F}_u^\dagger S^{-1}\mathbf{F}_m) + \mathbf{F}_m). \tag{56}$$



Jacobian

Now define the symmetric matrices

$$G \equiv \mathbf{F}_{\mathbf{m}}^{\dagger} S^{-1} \mathbf{F}_{\mathbf{m}}; \quad U \equiv (\mathbf{F}_{u}^{\dagger} S^{-1} \mathbf{F}_{u})^{-1}; \quad H \equiv \mathbf{F}_{m}^{\dagger} S^{-1} \mathbf{F}_{\mathbf{u}} \tag{57}$$

We have expressions for $\frac{\partial \lambda}{\partial m^0}$. Equation (53) is determined as:

$$\begin{aligned} J_{m,m^0} &= I - V(m^0)\mathbf{F}_{m}^{\dagger} \frac{\partial \lambda}{\partial m^0} = I - V\mathbf{F}_{m}^{\dagger}(-S^{-1}\mathbf{F}_{u}U^{-1}H^{\dagger} + S^{-1}\mathbf{F}_{m}) \\ &= I - V(G - HUH^{\dagger}) \end{aligned} \tag{58}$$

If we let $C = G - HUH^{\dagger}$, we obtain

$$V(m) = J_{m,m^0} V J_{m,m^0}^{\dagger} = V - 2VCV + VCVCV. \tag{59}$$

We would keep 2nd order term at the moment. Some materials like [1] had an error in this part.



Covariance of the Unknowns

Just like how we derived Eq.(59) we can estimate the variance matrix of the unknowns.

$$V_U = J_{u,m0} V J_{u,m0}^T \tag{60}$$

$J_{u,m0}$ can be obtained from Eq.(50). Defining

$$K \equiv ((\mathbf{F}_u^\dagger)^\nu S^{-1} \mathbf{F}_u^\nu)^{-1} (\mathbf{F}_u^\dagger)^\nu S^{-1} \tag{61}$$

we write:

$$J_{u^{\nu+1},m^0} = \frac{\partial u^{\nu+1}}{\partial m^0} = \frac{\partial u^\nu}{\partial m^0} - K \frac{\partial R}{\partial m^0} \simeq -K\mathbf{F}_m. \tag{62}$$

Note that we only have initial "Guess" for the unknowns; In principle, it is not a driven value from measurements. Then, $\frac{\partial u^0}{\partial m_0} = 0$. Also, we approximate that the terms in 2nd or higher iterations are negligible: $J_{u,m^0} \simeq J_{u^0,m^0}$.



Pull distribution

The covariance in Equation (22) is estimated as:

$$Cov(m, m^0) = J_{m,m^0} V(m) = V - VCV. \tag{63}$$

If we substitute this and Eq.(59) into Eq.(22), we get

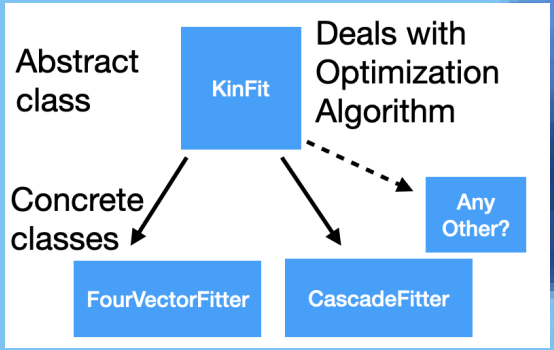
$$V(\epsilon) = VCVCV. \tag{64}$$

Note that 2nd order term affects the covariance matrix of the correction.



KinFit Package

$$\chi^2 = \delta M^T V^{-1} \delta M + 2\lambda f(M, U) \quad (65)$$



KinFIt provides...

- Minimize χ^2
- Calculate pulls and p-values

Users should...

- Assign proper variables for M and U
- Define physical constraints
- Write the derivatives, F_M and F_U by hand



References

[1] O. Skjeggestad Frodesen A.G. **Probability and Statistics in Particle Physics**. Columbia University Press, 1980. ISBN: 8200019063.

[2] B. Kang. **Bono-Beamer**. URL: https://github.com/kangbm94/Bono-Beamer.

